Polar and Cartesian Equation Conversions

Precalculus Challenge Practice — 30 Minutes

Name: _____ Date: _____

Instructions

Complete all 12 problems without a calculator. Show the conversion steps and use exact values. Problems 1–6 require Cartesian equations in x and y; Problems 7–12 require polar equations in r and θ. Simplify and include any necessary restrictions. Unless directed to solve for r, an equivalent factored polar equation is acceptable.

Allow signed values of r, with $0 \le \theta < 2\pi$. Preserve the entire curve, including the origin when it belongs to the original equation. If you divide by r, check the origin separately.

Reference

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad r^2 = x^2 + y^2.$$

$$\sin(2\theta) = 2\sin\theta\cos\theta, \qquad \cos(2\theta) = \cos^2\theta - \sin^2\theta.$$

For $r \neq 0$, $\cos\theta = x/r$ and $\sin\theta = y/r$. Also, $\sqrt{x^2+y^2} = |r|$, not necessarily r.

Polar to Cartesian

1. Convert $r = 6\cos\theta - 8\sin\theta$. Also state the center and radius of the resulting circle.

2. $r^2 = 3\cos\theta - 2\sin\theta$

3. $r^2 = 8\cos(2\theta)$

4. $r = \dfrac{4}{1+\sin\theta}$

Polar to Cartesian — Continued

5. $r = \dfrac{6}{3 - 2\cos\theta}$

6. $r^2 = \dfrac{12}{1 + \sin(2\theta)}$

Cartesian to Polar

7. Convert $(x-3)^2 + (y+2)^2 = 13$. Give an equation with r isolated.

8. Convert $x^2 + y^2 + 4x - 6y - 12 = 0$. Solve the resulting equation for r, including both real branches.

Cartesian to Polar — Continued

9. $x^2 - y^2 = 6$

10. $4x^2 + 9y^2 = 36$

11. Convert $(x^2 + y^2)^2 = 4x^2 + 12y^2$. Factor the resulting polar equation and state whether it includes the origin.

12. Convert to polar form and simplify:

$$(x^2 + y^2)^2 = 12xy.$$